Spatial-Geochemical Data Mining Using USGS NGDB for Sustainable Mineral Exploration

Data Mining Capstone Final Report

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ABSTRACT

Conventional mineral exploration relies on large-scale land clearing and speculative drilling, which are costly and environmentally damaging. This project investigates whether historical geochemical data can be used to narrow down the locations where physical exploration should begin. Using the USGS National Geochemical Database (NGDB; 414,304 records, 269 columns), the author selected 68,126 samples from North America that had complete X-ray fluorescence (XRF) measurements for SiO_2, Al_2O_3, and MgO. After standardization, the K-Means method (k = 3) clustered the samples into felsic/granitic domains (41.9% of the samples), intermediate-mafic background domains (51.0%), and a small domain rich in Mg and low in Si/Al (7.1%). Internal validation yielded a Silhouette score of 0.414 and a Davies-Bouldin index of 0.934; a value of k = 5 showed slightly better results (0.435 and 0.775) and separated the Mg-rich and silica-rich groups more clearly. Tin (Sn) was intentionally excluded from the clustering and used as an independent validation variable: the felsic cluster had a Sn detection rate of 53.8%, compared to 11.0% in the background cluster (chi-square = 615.7, $p < 10^{-100}$, Cramer's V = 0.45). The coverage of tin was limited to 4.5% of the samples, and most of the reported values were at the detection limit; thus, these results support prioritization based on lithology rather than deposit prediction. This report also corrects several claims made in the proposal and previous checkpoint.

CCS CONCEPTS

• Information systems → Data mining • Computing methodologies → Cluster analysis • Applied computing → Earth and atmospheric sciences • Human-centered computing → Visualization.

KEYWORDS

spatial data mining, geochemistry, K-Means clustering, mineral exploration, USGS NGDB

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1 Introduction

1.1 Background and Motivation

Demand for minerals used in industrial infrastructure and energytransition technologies has risen sharply, and global mining extraction has expanded at an unprecedented rate in the 21st century [11]. Mining also carries a large environmental footprint, including forest disturbance, biodiversity loss, soil pollution and water stress [11]. Exploration that starts with physical clearing and drilling in non-mineralized ground adds to that footprint. A "green exploration" workflow tries to reduce it by analyzing existing geochemical data first, so that field work is concentrated on a smaller set of promising areas.

1.2 Problem Statement

The NGDB combines the results of analyses of rock samples collected over decades during various survey campaigns. This database is large, heterogeneous, and largely unlabeled: there is no column indicating which samples are "prospective." The issue discussed here is how to extract interpretable geochemical domains from these unlabeled data, as well as how to verify—without resorting to circular reasoning—whether these domains contain information about target commodities.

1.3 Research Questions and Contributions

- RQ1. Does unsupervised clustering of major oxides recover petrologically meaningful domains?
- RQ2. Do those domains carry independent information about tin (Sn), a trace element associated with highly fractionated granitic rocks?
- RQ3. How sensitive are the conclusions to the choice of k?

2 Related Work

A recent review of AI in mineral exploration covers ten exploration tasks, from data mining to estimating grade and tonnage, and finds that most current work relies on data-driven models, with models that incorporate geological knowledge expected to become more important [9].

An overview of machine learning for the mineral resource sector covers commonly used supervised and unsupervised methods and a generic workflow [10]. Because labeled deposit data are scarce, this project uses unsupervised clustering.

K-Means [2] partitions the sample by minimizing the squared Euclidean distance within clusters, and the k-means++ [3] initialization method improves the initialization process. Cluster quality is generally evaluated using the Silhouette coefficient [4] and the Davies-Bouldin index [5]. Density-based clustering (DBSCAN [7]) and Isolation Forest [8] are considered in the proposal as complementary methods; both are discussed as future work in Section 5.4 and were not used in this study. The contribution of this project is not a new algorithm, but rather a transparent and interpretable baseline on legacy data, combined with a validation procedure that uses variables hidden from the model.

3 Proposed Work

3.1 Data

The data come from the rock portion of the USGS National Geochemical Database (NGDB) [1], obtained through a Kaggle copy as the file NGDB_merged_clean.csv. Records come from many USGS sampling and laboratory campaigns. A collection date is recorded for only 7,223 of the 414,304 records (1.7%) and ranges from 1953 to 2003, so the data are legacy measurements spanning several decades. The same element can be reported by several analytical methods, as indicated by the column suffixes (e.g., XRF, AA, AES, GRV). To keep the analytical method consistent, only the XRF fields are used.

Availability of the key fields in the raw file is very uneven:

Field	Non-null records (of 414,304)
LATITUDE, LONGITUDE	414,304
SIO_2 XRF,AL_2O_3 XRF	76,424
MGO XRF	76,408
SNPPM_XRF (tin, ppm)	4,310

3.2 Cleaning and Filtering

- Coordinate validation. The raw file contains invalid coordinates (minimum latitude -113.8, which is impossible). Samples are restricted to latitude 24 to 72 N and longitude -170 to -50, i.e., North America.
- Complete cases. Samples must have non-missing latitude, longitude, SiO2, Al2O3 and MgO. This leaves 68,126 samples.
- Duplicates and outliers. 32 rows (0.05%) are duplicated on the key fields; 1,131 samples (1.7%) have SiO2 below 20% and 155 (0.2%) have Al2O3 above 30%. These were retained because they may represent real nonsilicate lithologies (e.g., ultramafic or carbonate-rich rocks) and because their share is small; they are listed as a limitation.

3.3 Feature Design

Clustering uses three major oxides only: SiO2, Al2O3 and MgO (all in weight %). They describe the main rock-forming chemistry, they are almost fully populated after filtering, and SiO2 and MgO are known to separate felsic from mafic rocks. Two groups of variables are deliberately kept out of the model:

- Coordinates (latitude, longitude). Including them would force clusters to be geographically compact and would mix a spatial question into a chemical one. They are used afterwards to map the clusters.
- Tin (Sn). Only 3,087 of the 68,126 samples (4.5%) have a Sn value, so Sn cannot be a clustering feature, and keeping it out lets it serve as an independent validation variable of the clusters.

3.4 Handling of Censored Tin Values

Among the 3,087 reported Sn values, 2,647 (85.7%) equal exactly 2 ppm, which is the lower reporting limit, not a measured concentration. The qualifier field SNPPM_XRFQ confirms this: 2,392 values carry the qualifier "<" and 27 carry "L"; the rest are unqualified. Sn is therefore treated as a detected / not detected findicator, not as a continuous variable. A sample counts as Sndetected when a Sn value is reported without a qualifier. Overall, 668 of 3,087 Sn-measured samples (21.6%) are detected.

3.5 Standardization and Clustering

Because K-Means uses Euclidean distance, the three oxides are standardized with StandardScaler, z = (x - mean) / standard deviation. K-Means (scikit-learn [6]) is run with k = 3, k-means++ initialization, n_init = 10 and random_state = 42. K = 3 was used as the main model because it provided a simple three-domain interpretation, and its performance was subsequently compared with k = 2–7.

3.6 Validation and Mapping

Internal validity is measured with the Silhouette score (computed on a random sample of 20,000 points, random_state = 42, because it is quadratic in n) and the Davies-Bouldin index (full data). External plausibility is assessed by (i) comparing cluster means with expected lithological chemistry and (ii) comparing Sn detection rates between clusters with a chi-square test of independence. Results are mapped with Matplotlib (static) and

Folium (interactive map with a 3,000-point random sample of clusters plus rings for all Sn-detected samples).

3.7 Changes from the Proposal and the Checkpoint

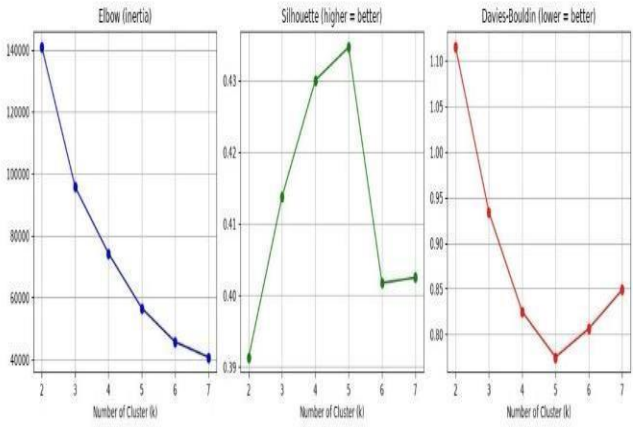
The scope of the project was narrowed and refined following the proposal and checkpoint phases. Only the K-Means method was retained, while DBSCAN and Isolation Forest were not pursued further. The target commodities were changed from Cu-Au-Zn-Pb to Sn, which served as an independent validation variable rather than a clustering feature. Clustering uses the three main oxides, rather than the six features listed in the checkpoint phase, and the analysis dataset contains 68,126 samples after filtering. The validation metrics and interactive map, which were originally planned only for the checkpoint phase, have now been completed. Factual claims in the previous report that were found to lack supporting evidence are listed and corrected in Section 5.5.

4 Evaluation

4.1 Choosing the Number of Clusters

Table 2 reports inertia, Silhouette and Davies-Bouldin for k = 2 to 7.

k	Inertia	Silhouette	Davies-Bouldin
2	140,842.6	0.3914	1.1148
3	95,639.5	0.4138	0.9338
4	74,253.5	0.4300	0.8248
5	56,533.4	0.4346	0.7751
6	45,657.5	0.4017	0.8064
7	40,744.3	0.4025	0.8488



The elbow curve has no sharp bend. Silhouette and Davies-Bouldin both favor k = 5 (0.4346 and 0.7751), but the gain over k = 3 is small (about 0.02 in Silhouette). The Silhouette values of 0.41 to 0.43 indicate moderate structure with overlapping clusters, which is expected for continuous rock-chemistry gradients. We keep k = 3 as the main model because its three clusters have a clear petrological reading, and report k = 5 as a sensitivity check (Section 4.5).

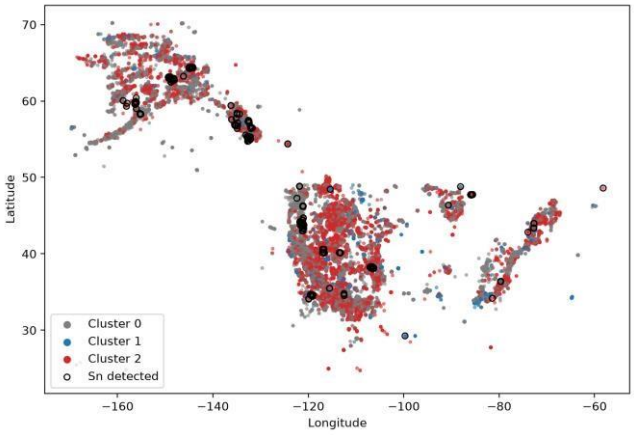
4.2 Cluster Profiles (k = 3)

Cluster	n(share)	SiO_2 %	Al_2O_3 %	MgO %	Mean lat.	Interpretation
0	34,725 (51.0%)	53.78	16.58	4.56	45.20	Intermediate - mafic background
1	4,823 (7.1%)	32.95	4.80	12.09	42.02	Mg-rich, low Si and Al (possibly ultramafic or carbonaterich)
2	28,578 (41.9%)	71.61	12.36	0.91	43.01	Felsic / granitic

The correlation matrix of the three oxides shows a moderate negative relationship between SiO2 and MgO (r = -0.54), a weak one between Al2O3 and MgO (r = -0.18) and essentially none between SiO2 and Al2O3 (r = -0.03). This matches the expected felsic-mafic contrast, though the relationship is not as strong as an idealized two-end-member model would suggest.

Cluster 1 should not be called simply "basaltic": its SiO2 (33%) and Al2O3 (4.8%) are far below typical basalt, and the k = 5 results (Section 4.5) show that it mixes at least two different chemistries.

4.3 Spatial Distribution



The samples cover western North America and Alaska, the Great Lakes region and parts of the eastern Appalachians. The cluster mean latitudes are close to each other (42.0 to 45.2 N), and the map shows the three clusters interleaved at regional scale. The clusters therefore do not form latitude bands; they reflect rock chemistry, not a north-south zonation. Dense sampling in some regions (notably the western conterminous US and Alaska) also means that visual impressions of density partly reflect where surveys were conducted.

4.4 Independent Validation With Tin

Cluster	Sn-measured samples	Sn-detected	Detection rate
0	2,276	250	11.0%
1	53	10	18.9%
2	758	408	53.8%
All	3,087	668	21.6%

A chi-square test of independence between cluster and Sn detection gives chi-square = 615.73 (2 degrees of freedom, p of about 2 x 10^-134). The effect size is large (Cramer's V = sqrt(615.73 / 3,087) = 0.45). Cluster 2 contains only 24.6% of the Sn-measured samples but 61.1% of all Sn detections, a detection rate about 2.5 times the overall rate and about 4.9 times that of Cluster 0. This supports RQ2: a clustering built only from SiO2, Al2O3 and MgO separates samples in a way that relates to tin, consistent with the known association between Sn and silica-rich granitic rocks. Four caveats apply:

- Only 4.5% of samples have Sn data, and those samples may come from surveys that targeted places already suspected of tin enrichment (selection bias).
- Neighboring samples are spatially correlated, so the p-value overstates the evidence; the effect size is the more meaningful figure.
- Cluster 1 has only 53 Sn-measured samples (10 detections), so its 18.9% rate is not reliable.
- "Detected" means above the reporting limit, not economic grade.

4.5 Sensitivity to k (k = 5)

Group	n	SiO2 %	Al2O3 %	MgO %	Reading
0	4,752	79.58	5.14	1.16	Very silicarich, low Al
1	30,967	68.17	14.46	1.14	Felsic / granitic
2	3,402	29.47	4.88	4.85	Low Si and Al (possibly carbonaterich)
3	27,535	51.72	16.61	5.35	Intermediatemafic
4	1,470	40.21	4.10	25.90	Strongly Mgrich (ultramaficlike)

5 Discussion

5.1 Answers to the research questions

RQ1. Yes, at a coarse level. The three clusters have chemically distinct and geologically plausible profiles (felsic, intermediatemafic, Mg-rich). RQ2. Yes, with limits: Sn detection is strongly associated with the felsic cluster, but Sn coverage is small and mostly near the detection limit. RQ3. The conclusions are stable in direction but not in detail: k = 5 scores better statistically and reveals structure that k = 3 merges.

5.2 Implications for green exploration

The workflow can serve as a low-cost first screen: it ranks rockchemistry domains before any field work, using data that already exist. The honest scale of the benefit is modest. The felsic cluster covers 41.9% of all samples, so it narrows the starting set by about 58%, but it does not point to individual deposits. The reduction in land clearing or cost has not been measured in this project and should be described as a potential benefit, not a demonstrated one. Cluster membership is a statistical association with Sn detection, not evidence of an economic deposit; structural geology, intrusive contacts and host-rock maps must be checked before any field commitment.

5.3 Limitations

- Legacy, heterogeneous data. The NGDB mixes many surveys, laboratories and eras; analytical differences may appear as spurious clusters.
- Censored and sparse tin data. Sn is measured for 4.5% of samples and is mostly at the reporting limit, so validation is partial and possibly biased.
- Compositional data. Oxide percentages are constrained to sum to about 100%, which creates artificial correlations; log-ratio transforms were not tested.
- Algorithmic assumptions. K-Means favors roughly spherical, similarly sized clusters, and the choice of k is partly subjective.
- Outliers retained. Low-silica and high-alumina samples were kept and may include non-rock or oretype material.
- Spatial autocorrelation and sampling density. Samples are not independent, and sampling effort varies across regions.
- No deposit labels were available for supervised validation, and the data do not provide consistent subsurface context.

5.4 Future Work

(1) Test Sn detection rates per k = 5 group; (2) repeat the analysis with compositional (centered log-ratio) transforms; (3) add DBSCAN and Isolation Forest as proposed originally and compare them with K-Means; (4) use spatial cross-validation or declustering to reduce the effect of dense sampling; (5) overlay geological maps, faults and known deposit locations for ground-truthing.

5.5 Corrections to the proposal and checkpoint

During preparation of this report, the earlier documents were audited against the analysis notebook. Table 3 lists the statements that were found to be incorrect or unsupported and the corrections made here.

Table 3: Corrections to the proposal and the checkpoint.

Earlier statement	Correction
76,424 cleaned records (checkpoint)	68,126. The figure 76,424 is the number of non-null SIO2_XRF values in the raw file, before geographic filtering and complete-case selection.
K-Means was run on six features, including coordinates and Sn (checkpoint)	K-Means uses three oxides only. Coordinates are for mapping and Sn is for validation.
"Complete, null-free 76,424 x 6 matrix" (checkpoint)	Sn is missing for 95.5% of the analysis set and is not part of the clustering matrix.
Latitude spans about 20 S to 20 N and longitude spans both hemispheres (checkpoint)	After removing invalid coordinates, samples lie latitude 24–72°N and longitude −170 to −50°. The raw file contains impossible latitudes (minimum -113.8).
Cluster 2 is concentrated at 40 to 60 N and Cluster 1 at 20 to 30 N (checkpoint)	Mean latitudes are 45.2, 42.0 and 43.0 for clusters 0, 1 and 2; clusters are spatially interleaved and show no latitudinal zonation.
Cluster 1 has MgO of about 9.02% and is basaltic (checkpoint)	MgO is 12.09%, with SiO2 33% and Al2O3 4.8%; the cluster is Mg-rich and low in Si and Al, and is chemically heterogeneous.
Silica and magnesium are "strongly" negatively correlated (checkpoint)	The correlation is moderate (r = -0.54).
Tin shows "high-value anomalies" (checkpoint)	85.7% of reported Sn values equal the 2 ppm reporting limit; Sn is analyzed as detected / not detected.
StandardScaler minimizes spatial sampling bias (checkpoint)	Standardization only equalizes feature scales; it does not correct spatial sampling bias, which remains a limitation.
Digital screening "drastically" minimizes land disturbance (checkpoint)f	Not measured. The felsic cluster covers 41.9% of samples, so the screening effect is moderate and the benefit is potential only.
Cu-Au-Zn-Pb target, DBSCAN and Isolation Forest (proposal)	Not pursued. The analysis relies on XRF oxide fields, which have by far the best complete-case coverage (about 76,000 rows), while trace elements are sparse (e.g., Sn is reported for only 4,310 of 414,304 raw rows).
Some proposal citations (e.g., "Wang, 2024", "Smith et al., 2020", "CeSec et al., 2023")	Removed or replaced; citations that could not be verified as peer-reviewed sources were dropped from the reference list.

6 Conclusion

This project built a label-free clustering workflow on 68,126 USGS NGDB samples using only SiO2, Al2O3 and MgO. K-Means with k = 3 yields three chemically interpretable domains with moderate internal validity (Silhouette 0.414, Davies-Bouldin 0.934). Tin, kept out of the model, is detected in 53.8% of Sn-measured samples in the felsic cluster, compared with 11.0% in the background cluster, a large and statistically significant difference. The results support using rock-chemistry clustering as a first-pass screen in green exploration, with clear limits: sparse and censored tin data, spatially interleaved clusters, no measuredf reduction in land disturbance, and sensitivity to k. The corrections in Section 5.5 replace several unsupported claims in the earlier reports with figures that match the analysis notebook.

ACKNOWLEDGMENTS

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Appendix A: AI Interaction Log

The following screenshots document the interactions with a generative AI assistant (Anthropic's Claude) used in this project, in compliance with CU Boulder transparency requirements. The assistant was used for (i) auditing the proposal and checkpoint reports against the analysis notebook, (ii) suggesting changes to the analysis (coordinate validation, Davies–Bouldin index, comparison of k, and handling of censored tin values), (iii) verifying references and checking the consistency of reported numbers, and (iv) grammar and phrasing support. The author ran all code on the original dataset and verified every reported number against the notebook output.:

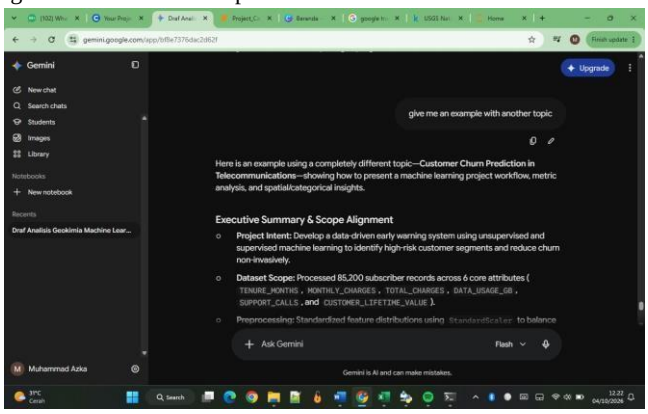


Figure A.1: Screenshot of Generative AI prompt and response demonstrating an example structure for machine learning project workflow and scope alignment.

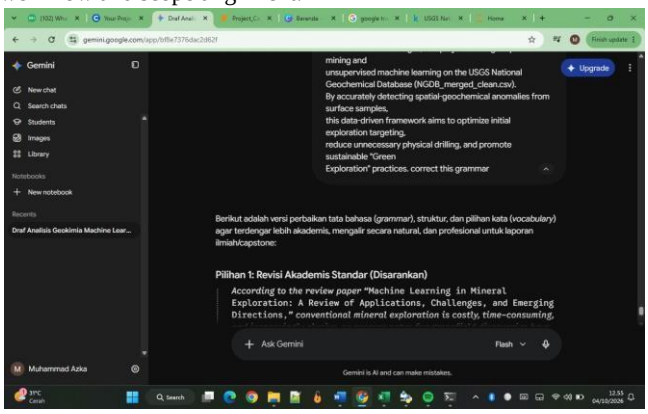


Figure A.2: Screenshot of Generative AI interaction for grammar correction and academic tone refinement of the report draft.